

Recursively defined Sets and Structures

we are going to look at this because we want to use induction on structures and sets. (Trying to extend induction beyond natural numbers)

Def

A recursively defined structure/set is a structure/set S defined by:

1. Its smallest element(s) (base case(s))
2. Rule(s) that construct elements out of smaller elements.
3. The structure/set includes only elements that can be constructed from the base case and the recursive case.

$$S = \left\{ x : x \text{ is either case 1 or } x \text{ follows case 2} \right\}$$

Ex: The set of non-negative integers

1. 0
2. If k is a non-negative integer, $k+1$ is also a non-negative integer.

Smallest element
that we can
generate is

0.

I can make 1 using 0
and applying rule (2)

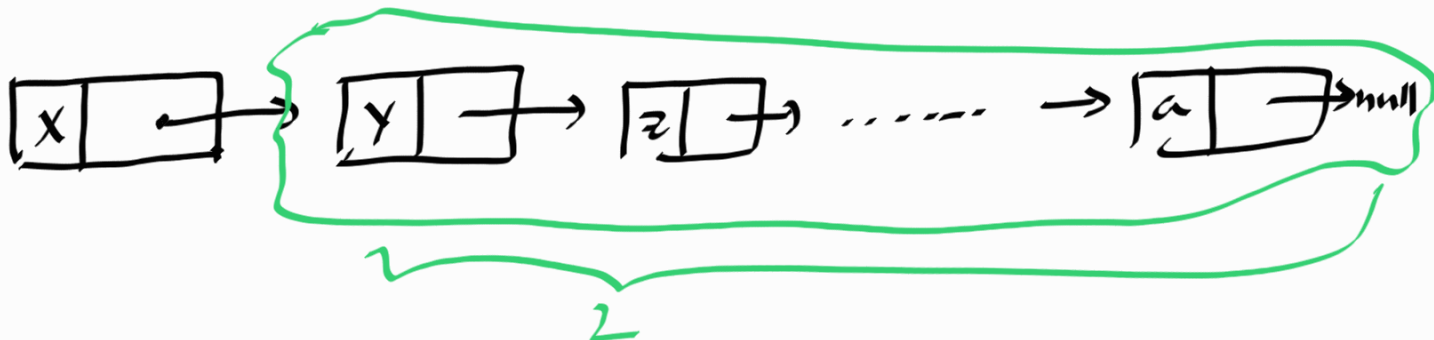
$$0+1=1$$

1 is a non-negative integer.

Ex: A recursively defined structure.

Let's try to define Linked lists as a
recursively defined structure.

1. Base case: Empty linked list $[]$, null
2. Recursive case: $[x, L]$, where x is some data and L is a linked list



smallest linked list

$[]$, null

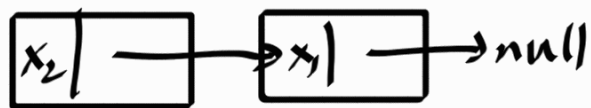
Linked list with 1-element

$[x, []]$

$\boxed{x} \rightarrow \text{null}$

Linked list with 2-elements

$[x_2, [x_1, []]]$



Ex: set of well formed statements of propositional logic over a set of boolean variables X .

$$X = \{t, q, r\}$$

$$t \Rightarrow q, q \wedge r, r \vee t, \neg r, \neg t, (q \vee r) \Rightarrow (r \vee t)$$

t, q, r

1. P , for some $P \in X$

2

2a) If P, S are well formed sentences of propositional logic;

$$P \star S \text{ where } \star \in \{\wedge, \vee, \Rightarrow, \Leftrightarrow, \oplus\}$$

2b) If P is a wfs of propositional logic, $\neg P$ is a well formed statement of propositional logic.

$$X = \{a, b, c, d\}$$

base case elements

a, b, c, d

recursive cases

$a \vee b, a \wedge b$

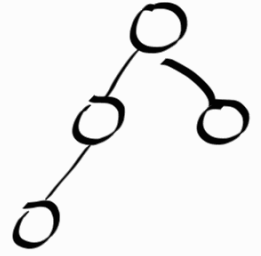
$a \neg (a \vee b)$

$a \Rightarrow b$

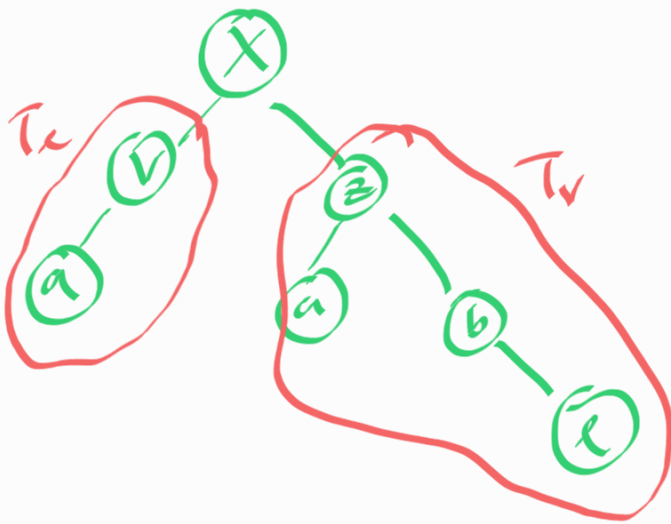
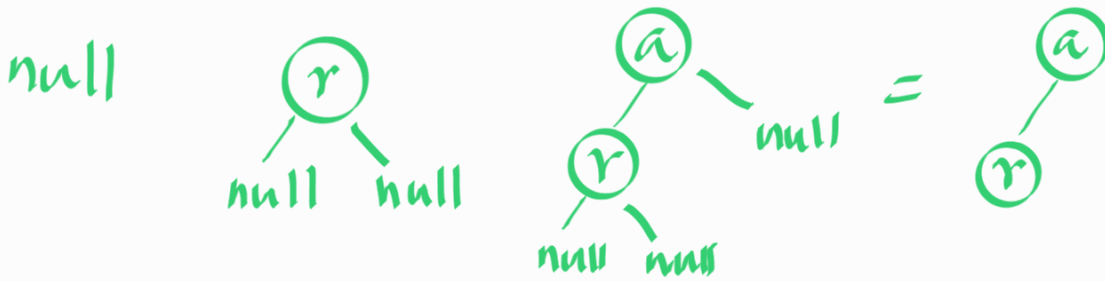
$$S = \{a, b, c, d, a \vee b, a \wedge b, a \oplus b, \dots\}$$

Ex: Binary trees as recursively defined structure.

1. null (empty tree) —



2. root node r , left subtree T_L , right subtree T_R where r is an arbitrary value, & T_L, T_R are both binary trees.



Def: Suppose you want to prove $[\forall n \geq 0 : P(n)]$. To give a proof using mathematical induction we prove the following

1. Base case ($P(0)$)
2. Inductive case $[\forall n \geq 1 : P(n-1) \Rightarrow P(n)]$

$$[\forall n \geq 0 : P(n)]$$

weak induction

Def: Strong induction

Suppose you want to prove $P(n)$ holds for $n \geq 0$. To give a proof by strong induction for $[\forall n \geq 0 : P(n)]$, we need to prove the following:

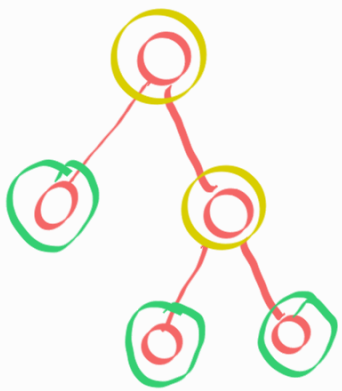
1. Base case ($P(0)$)
2. Inductive case:

$$\forall n \geq 1 : [P(0) \wedge P(1) \wedge P(2) \dots \wedge P(n-1) \Rightarrow P(n)]$$

Both induction techniques are equivalent.

Terms involving binary trees.

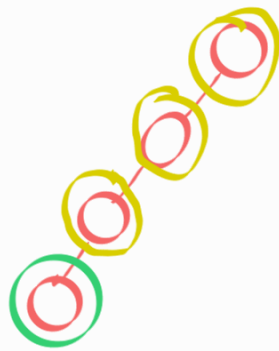
- Binary tree is binary because each node has at most 2 children.
- Each edge connects pairs of nodes.
- Node is a leaf node if it has null for both children.
- Node is an internal node, if it is not a leaf node.



leaf node : 3

internal nodes : 2

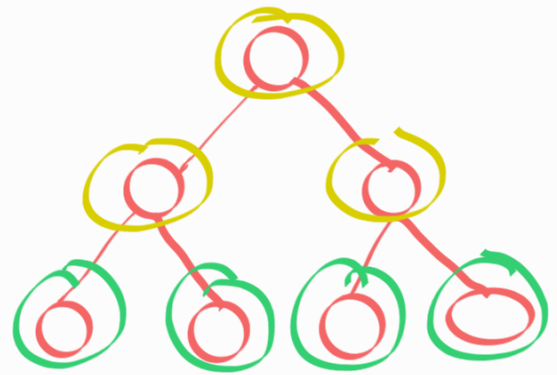
$$3 \leq 2 + 1 \quad \checkmark$$



leaf node : 1

internal nodes : 3

$$1 \leq 3 + 1 \quad \checkmark$$



leaf node : 4

internal nodes : 3

$$4 \leq 3 + 1 \quad \checkmark$$

Claim: In any binary tree T ;

$$\text{leaves}(T) \leq \text{internals}(T) + 1$$

we will try to prove this using strong induction.

Proof:

step 01: Let T be binary tree created by using rule (2) n # of times.

$P: \mathbb{N} \rightarrow \{T, F\}$, $P(n)$ is defined as follows:

$$P(n) = \begin{cases} T, & \text{If } \text{leaves}(T) \leq \text{internals}(T) + 1 \\ F, & \text{otherwise} \end{cases}$$

$P(n)$ is true, if for any binary tree T constructed by n applications of rule 2 follows the inequality $\text{leaves}(T) \leq \text{internals}(T) + 1$.

step 02: state the variable of which you are performing induction
variable $n \in \mathbb{N}$

step 03: state the base case
 $n=0$

Step 04'. prove the base case.

The binary tree T constructed using 0 number of applications of rule 2 is the empty tree, and this tree is unique.

for empty tree T ,

$$\text{leaves}(T) = 0 \quad \text{internals}(T) = 0$$

$$0 \leq 0 + 1$$

Base is proven ($P(0)$)

Step 05'. state the inductive step

$$\forall n \geq 1, P(0) \wedge P(1) \wedge P(2) \wedge \dots \wedge P(n-1) \Rightarrow P(n)$$

If any binary tree B generated using $k \leq n$ # of rule 2 applications has $\text{leaves}(B) \leq \text{internals}(B) + 1$, then the binary tree T generated using n rule 2 applications must follow $\text{leaves}(T) \leq \text{internals}(T) + 1$

Step 06: proof of inductive step

WTS: $\{ \forall n \geq 1: P(n) \wedge P(n+1) \dots \wedge P(n-1) \Rightarrow P(n) \}$

we will consider 2 cases:



case 1: $n=1$

case 2: $n \geq 2$

case 1: The only way to create binary tree T with one application of rule 2 is to use rule 1 for both sub trees. Hence, the tree T must only contain one node.

T contains 1 leaf node

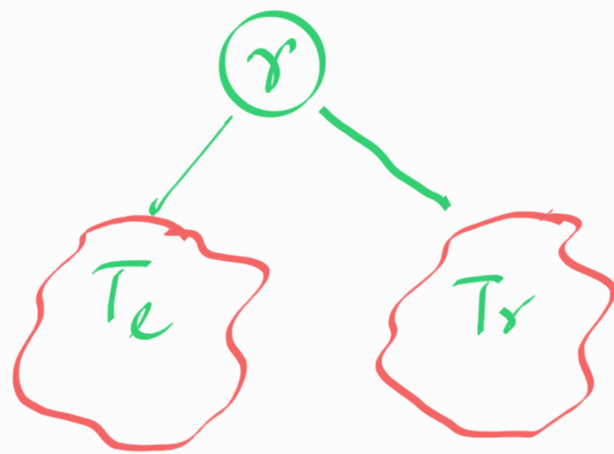
T contains 0 internal nodes



$$1 \leq 0 + 1 \quad \checkmark$$

case 2: when $n \geq 2$, observe that tree must be generated using "left subtree T_l using $l \neq$ of rule 2 applications and a right subtree T_r

created using r # of rule 2 applications. Then we need to use rule 2 one more time to tie them together.



Therefore, for T , $n = l + r + l$, $l < n, r < n$

$\forall n \geq 1: P(0) \wedge P(1) \wedge \dots \wedge P(n-1) \Rightarrow P(n)$

Assume $P(0) \wedge P(1) \wedge \dots \wedge P(n-1)$ is true
(Inductive hypothesis)

Now we are going to apply inductive hypothesis:

$$\text{leaves}(T_e) \leq \text{internals}(T_e) + 1$$

$$\text{leaves}(T_r) \leq \text{internals}(T_r) + 1$$

Also, observe that either $T_e \neq \text{null}$ or $T_r \neq \text{null}$ or both of them are not equal to null

Therefore, leaves of T are leaves of T_L and T_R , and internals of T are internals of T_L, T_R and the root node.

$$\begin{aligned} \text{leaves}(T) &= \text{leaves}(T_L) + \text{leaves}(T_R) \\ \text{internals}(T) &= \text{internals}(T_L) + \text{internals}(T_R) + 1 \\ \rightarrow \text{leaves}(T) &= \text{leaves}(T_L) + \text{leaves}(T_R) \end{aligned}$$

$$\text{leaves}(T) \leq \text{internals}(T_L) + 1 + \text{internals}(T_R) + 1$$

$$\leq \underbrace{\text{internals}(T_L) + \text{internals}(T_R) + 1}_{\text{internals}(T)} + 1$$

$$\text{leaves}(T) \leq \text{internals}(T) + 1$$

$$\underbrace{\hspace{10em}}_{P(n)}$$

Step 0+! We have proven the base case and strong inductive case
 $\therefore$ using principles of mathematical induction
 $[\forall n \geq 0: P(n)] //$

Popup Quiz 08

Prove

$\forall n \geq 0: n^3 + 2n$ is divisible by 3.

$$P: \mathbb{N} \rightarrow \{T, F\}$$

$$P(n) = \begin{cases} T & , \text{ if } 3 \mid n^3 + 2n \\ F & , \text{ otherwise} \end{cases}$$

base case $P(0)$

when $n=0$

$$0^3 + 2 \cdot 0 = 0$$

$$3 \mid 0 \quad \checkmark$$

$P(0)$

inductive case

$$\forall n \geq 1: P(n-1) \Rightarrow P(n)$$

Assume $P(n-1)$

$$3 \mid (n-1)^3 + 2(n-1) \quad (\text{Inductive hypothesis})$$

WTS! $P(n)$

$$3 \mid n^3 + 2n$$

consider $n^3 + 2n$

$$(n-1+1)^3 + 2(n-1+1)$$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(n-1)^3 + 3(n-1)^2 \cdot 1 + 3(n-1) \cdot 1^2 + 1^3 + 2(n-1) + 2$$

$$\underbrace{(n-1)^3 + 2(n-1)}_{3 \cdot k} + 3(n-1)^2 + 3$$

$$3 \left(\underbrace{k + (n-1)^2 + 1}_{c \in \mathbb{Z}} \right)$$

$$3 \mid n^3 + 2n$$